

Robot Vacuum - Engineering BOM

Reverse Engineering Documentation

February 14, 2026

Classification Legend

- **Project Code:**
- **Group Codes (Mnemonic):**
 - **C:** Chassis - *Structural Base*
 - **M:** Mechanics - *Moving Parts*
 - **E:** Electronics - *Brain & Power*
 - **U:** User/Shell - *External Housing*
 - **H:** Hardware - *Fasteners & Springs*
- **ASM (Assembly):** Sub-assembly containing multiple parts.
- **PRT (Part):** Single, non-demountable component.

1 Group C: Chassis (Structural Skeleton)

The foundation: Internal frames, housings, and fixed plastic parts.

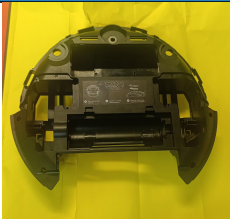
Image	Part ID	Part Name	Type	Notes
	C01	Main Chassis (Bottom Case)	PRT	Main plastic body frame.
	C02	Base Plate (Disk)	PRT	Circular bottom cover.

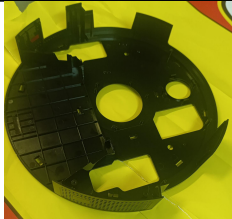
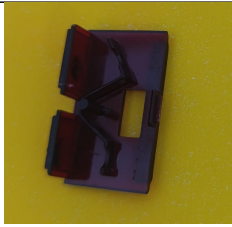
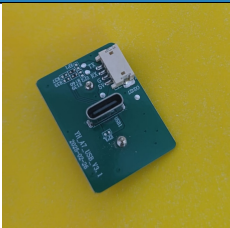
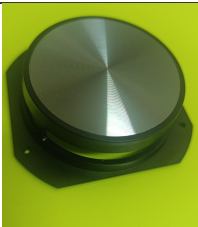
Image	Part ID	Part Name	Type	Notes
	C03-01	Inner Frame	PRT	Internal structural support.
	C03-02	IR window	PRT	Mini IR Window 2pcs
	C04	Board Holder Bracket	PRT	Black plastic support bracket.
	C05-01	Wall Sensor Housing	PRT	Black plastic sensor enclosure. *4pcs
	C05-02	Wall Sensor IR Window	PRT	Black plastic sensor window. *4pcs
	C05-03	Wall Sensor Board	PRT	Sensor board. *4pcs
	C06	Battery/Service Door	PRT	Cover with model label.
	C07-01	Type-C Port Cover	PRT	Charging port plastic housing.

Image	Part ID	Part Name	Type	Notes
	C07-02	Type-C Port Board	PRT	Charging port board.
	C08	LIDAR Top Cap	PRT	Cosmetic cover for laser (Moved from Shell).
	C09	Bumper Frame	PRT	External shock absorber.
	C10-01	Bumper IR Window	PRT	Dark transparent plastic.
	C10-02	Bumper Lower Trim	PRT	Rubber guard/seal for bumper bottom.
	C10-03	Bumper IR Window	PRT	Dark transparent curved plastic.

2 Group M: Mechanics (Actuation & Muscles)

Image	Part ID	Part Name	Type	Notes
	M01L	Drive Wheel Module (L/R)	ASM	DC motor + gearbox + tire.

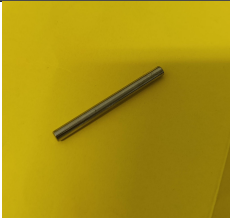
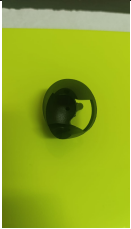
Image	Part ID	Part Name	Type	Notes
	M01R	Drive Wheel Module (L/R)	ASM	DC motor + gearbox + tire.
	M01R-01	Right Wheel Gearbox Housing	PRT	Plastic gear housing.
	M01R-02	Right Wheel DC Motor	PRT	Brushed DC motor with pinion.
	M01R-03	Right Wheel Gearbox Cover	PRT	Plastic cover for gears.
	M01R-04	Steel pin of gearbox	PRT	Steel pin for hinge.
	M02	Front Caster Housing	PRT	Swivel housing.
	M03	Front Caster Roller	PRT	Black/White roller ball.
	M04	Vacuum Fan (EBO)	ASM	Centrifugal fan blower.

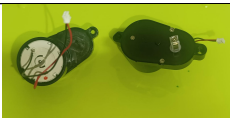
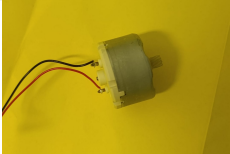
Image	Part ID	Part Name	Type	Notes
	M04-01	Blower Motor & Impeller	ASM	EBO DC motor assembly.
	M04-02	Blower Housing (Volute)	PRT	Plastic casing for vacuum fan.
	M04-03	Blower Exhaust Gasket	PRT	Rubber vibration/seal pad.
	M04-04	Rubber Pins	PRT	Blower Housing Rubber pins.
	M05	Side Brush Motor Box	ASM	Geared motor assembly.
	M05-01	Side Brush Gearbox	PRT	Gear reduction housing.
	M05-02	Side Brush DC Motor	PRT	Brushed motor with worm gear.
	M06L	Secondary Actuator	ASM	Small DC motor unit.
	M06R	Secondary Actuator	ASM	Small DC motor unit.

Image	Part ID	Part Name	Type	Notes
	M06R-01	Actuator Gearbox Housing	PRT	Right side small gear housing.
	M06R-02	Actuator DC Motor	PRT	Small brushed motor with pinion.

3 Group E: Electronics (Brain, Sensors & Power)

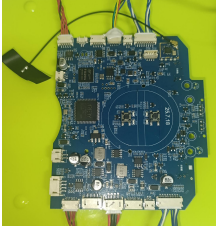
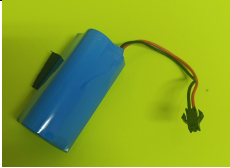
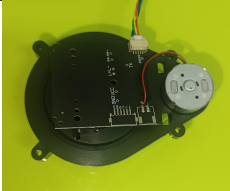
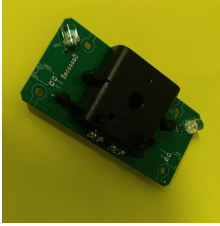
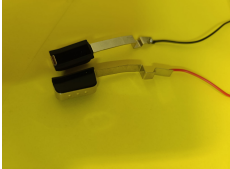
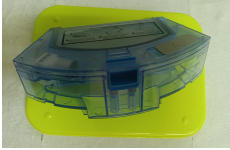
Image	Part ID	Part Name	Type	Notes
	E01	Mainboard PCBA	ASM	Central processing unit.
	E02	Li-ion Battery Pack	ASM	Blue shrink wrap, 2600mAh.
	E03	LIDAR Module	ASM	Laser Distance Sensor unit.
	E04	UI Button Module	ASM	Power/Home interface PCB.
	E05	Edge/Cliff Sensor PCB	ASM	Green sensor boards.

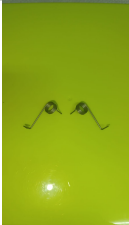
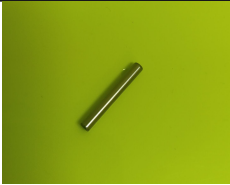
Image	Part ID	Part Name	Type	Notes
	E06	Speaker Unit	ASM	Audio output speaker.
	E07	Charging Dock Station	ASM	External power base station.
	E07-01	Dock Top Housing	PRT	Upper plastic shell of the dock.
	E07-02	Dock IR Window	PRT	Dark red transparent cover for IR alignment.
	E07-03	Dock Base Housing	PRT	Lower plastic base with mounting points.
	E07-04	Dock Mainboard PCBA	ASM	Circuit board with IR emitters.
	E07-05	Dock Charging Contacts	PRT	Metal spring plates with wire leads.

4 Group U: User/Shell (External & Consumables)

Image	Part ID	Part Name	Type	Notes
	U01	Hybrid Dust/Water Tank	ASM	2-in-1 container.
	U02	Filter Frame	PRT	HEPA filter holder.
	U03	Main Roller Brush	PRT	Blue brush.*2pcs bristled
	U04	Side Brush Pair	PRT	3-arm sweeping brushes.
	U05	Mop Attachment Bracket	PRT	Base plate for mopping cloth.

5 Group H: Hardware (Springs, Fasteners & Misc)

Image	Part ID	Part Name	Type	Notes
	H01	Metal Cable Plates	PRT	Shielding/retention plates.
	H02	Leaf Spring Contact	PRT	Bended metal plate.

Image	Part ID	Part Name	Type	Notes
	H03	Extension Springs	PRT	Wheel suspension springs.
	H04	Torsion Spring Pair	PRT	Mechanism springs.
	H05	Locking Latches	PRT	Black plastic retention hooks.
	H06	Screw Set A (Yellow)	PRT	Mixed assembly screws.
	H07	Screw Set B (White)	PRT	Additional fasteners.
	H08L	UI Button Plastics	PRT	Left Button.
	H08R	UI Button Plastics	PRT	Right Button.
	H09	Steel Pivot Pin	PRT	Metal shaft.